Juan Manuel Serena Ceceña

DATA ANALYST | JUNIOR DATA SCIENTIST | PYTHON & MACHINE LEARNING

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GitHub: github.com/JuanSeCe1402 | LinkedIn: Juan Manuel Serena Ceceña | Project portfolio

PROFESSIONAL SUMMARY

Data Analyst / Junior Data Scientist candidate with a physics background and hands-on work in Python, SQL and machine learning. Developed data-cleaning and preparation workflows, time-based prediction experiments and scientific computing applications. Focused on reproducibility, data quality and careful interpretation of model results.

TECHNICAL SKILLS

Python, SQL, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, Matplotlib; data cleaning and profiling, statistical analysis, classification, model evaluation, temporal validation; SQLite, MySQL, Flask, FastAPI, Git/GitHub and automated testing.

SELECTED PROJECTS

DARWIN - Customer repurchase prediction | Machine Learning / 2026 | View project

- Prepared UCI Online Retail II transactions for 30-day repurchase prediction using 180 days of purchase history and time-based train/validation/test splits (2,772 customers in the final test).
- Compared supervised models and baselines; Random Forest reached 0.608 average precision vs. 0.428 for a recency baseline on one retrospective holdout. Not a measured sales uplift.

NANA - CSV cleaning and data profiling | Data Engineering / 2026 | View project

- Built a pandas workflow for header normalization, missing values and duplicate handling, column profiling, identifier preservation and CSV/SQLite export; 30 locally passed test cases.

Bioimpedance - Simulation and parameter inference | Scientific ML / Research | View project

- Explored synthetic-data simulation and Ridge/PyTorch models with separate train/validation/test sets and training-only normalization; no clinical or experimental validation claimed.

ESCULAPIO - Scheduling and data operations | Python / SQL / 2026 | View project

- Built a WUNSCE administrative application using FastAPI, SQLAlchemy and SQLite for appointments, access roles and concurrency tests; demo uses fictitious records and no live integrations.

RESEARCH EXPERIENCE

Academic research participation at Universidad Autónoma de Nuevo León (UANL) across space weather, nanomaterials, Tlapiani Observatory and photonics labs. Worked with numerical modeling, cloud and atmospheric data analysis, scientific simulations and interpretation of research outputs.

EDUCATION

Universidad Autónoma de Nuevo León (UANL) | Physics studies - degree completion pending

LANGUAGES AND LEADERSHIP

Spanish: native | English: advanced | French: intermediate (DELF B1). Vice President, UANL Physics Students Association (2022-2023).